Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your latest score.

Next item →

Instructions

1. Which of the following elements are essential structural components of the Portable Executable (PE) file format, fundamental to its function within Windows operating systems? (Select all that apply) — 1 / 1 point

- [x] Section Table

  Nice work

  Correct! The Section Table provides detailed information about the layout and content of each section within the PE file, playing a pivotal role in its functionality.

- [x] DOS Header

  Nice work

  Correct! The DOS Header is crucial for ensuring backward compatibility with legacy systems, serving as a starting point for the PE file.

- [ ] PE Footer
- [x] NT Headers

  Nice work

  Correct! The NT Headers contain critical metadata, including the file signature, optional header, and file characteristics, making them integral to the PE structure.

- [ ] ELF Header

2. In the context of machine learning-based malware detection, which of the following features derived from PE files are most relevant for training algorithms to distinguish between benign and malicious software? (Select all that apply) — 1 / 1 point

- [x] Section names

  Nice work

  Correct! Section names can reveal abnormal structures or manipulation within the PE file, providing critical insights for machine learning models in malware detection.

- [x] Import table functions

  Nice work

  Correct! The functions listed in the import table reflect the external libraries the application depends on, providing insights into the program's behavior, which can be exploited to identify malicious activity.

- [ ] Author's name
- [x] Byte entropy

  Nice work

  Correct! Byte entropy is a valuable feature as it measures the randomness or predictability of byte sequences, often signaling obfuscation techniques in malware.

- [ ] File creation date

3. A security analyst encounters a suspicious PE file while performing a malware analysis. They notice that the file's .text section has an unusually high byte entropy and contains a large number of unfamiliar API calls. What could these characteristics indicate, and what might be the next logical step in the analysis process? — 1 / 1 point

- ( ) The file is attempting to evade detection by antivirus software, so static signature analysis should be sufficient to confirm its nature.
- ( ) The file is likely corrupted due to the presence of unfamiliar API calls, requiring manual decryption of the `.text` section.
- (•) The high byte entropy suggests potential obfuscation or encryption, and the unfamiliar API calls may indicate malicious behavior. Dynamic analysis in a controlled environment is needed to observe the file's behavior and assess the potential threat.
- ( ) The characteristics suggest the file is benign, as high entropy is typical of standard encryption methods, so no further analysis is required.

  Nice work

  Correct! High byte entropy often points to obfuscation or encryption, while unfamiliar API calls may signal malicious intent. Dynamic analysis is crucial to understand the executable's true behavior in a safe environment.

4. What is the primary operational function of the Import Address Table (IAT) in a PE file, particularly in relation to external dependencies? — 1 / 1 point

- (•) To dynamically resolve addresses of imported functions at runtime, ensuring correct function calls to external libraries
- ( ) To allocate memory required for the application during execution
- ( ) To store the application's executable instructions
- ( ) To maintain debugging information for use during development or analysis

  Nice work

  Correct! The IAT is responsible for dynamically linking external functions from libraries such as DLLs at runtime, allowing the executable to properly utilize external resources.

5. What is the role of the `.rsrc` section within a Portable Executable (PE) file, and why is it significant during malware analysis? — 1 / 1 point

- ( ) It holds the executable code for the program's core functionality.
- ( ) It stores detailed debugging information to aid in troubleshooting or reverse engineering.
- ( ) It manages memory allocation for the application's runtime operations.
- (•) It stores embedded resources such as icons, bitmaps, and strings required by the application, which may also be exploited for malicious purposes.

  Nice work

  Correct! The `.rsrc` section contains resources that the application utilizes during execution, and it is significant in malware analysis because attackers often embed harmful payloads or disguise malicious content within these resources.